Research Statement

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September 2026

1 Summary of Research Interests

My principal field of study is analysis, with focus on spectral geometry, fourier integral operators, and microlocal analysis. I am interested in expanding into other fields, in particular analytic number theory and stochastic differential equations (SDE).

2 Research Synopsis

2.1 Historical Background

In 1911, Hermann Weyl proved that given a Laplace-Beltrami operator, Δ_g, compact Riemmanian manifold (M, g) satisfies

$$\lim_{\lambda\to\infty} \frac{N(\lambda)}{\lambda^n} = (2\pi)^{-n}\omega_n \mathrm{Vol}_g(M)$$

where

$$N(\lambda) = \{\lambda_j^2 \leq \lambda \mid \lambda_j \text{ is an eigenvalue of } \Delta_g\},$$

$n = \dim M$, and ω_n is the volume of the unit ball in $\mathbb{R}^n$. Since then there have been a number of refinements and generalizations. Most notably, Duistermaat and Guillemin [DG75] proved that

$$N(\lambda) = (2\pi)^{-n}\omega_n \mathrm{Vol}_g(M)\lambda^n + o(\lambda^{n-1})$$

under certain dynamical conditions (specifically that the set of periodic bicharacteristics has measure 0).

In another direction, one may consider weighted counting functions of the type

$$N_A(\lambda) = \sum_{\lambda_j\leq\lambda} ||Ae_j||_{L^2(\Sigma)}$$

where $A : M \to \Sigma$ is a fourier integral operator. In particular, Zelditch [Zel92] proved that the counting function

$$N_H(\lambda) = \sum_{\lambda_j\leq\lambda} ||e_j||_{L^2(M)}$$

where H is an embedded Riemmanian submanifold of (M, g) satisfies

$$N_H(\lambda) = C_{M,H}\lambda^{n-d} + O(\lambda^{n-d-1})$$

d the dimension of H and $C_{M,H} = (2\pi)^{n-d}\omega_{n-d}\operatorname{Vol}_g(H)$. This asymptotic is analogous to the Kuznetsov Formula from analytic number theory so I will refer to it as the Kuznetsov Formula for Laplace–Beltrami Operators.

2.2 Refinements to the Kuznetsov Formula for Laplace–Beltrami Operators

Wyman and Xi (2023) [WX23] were able refine the lower order terms of Zelditch's formula and prove that

$$N_H(\lambda) = C_{M,H}\lambda^{n-d} + C + o(\lambda^{n-d-1})$$

under dynamical constraints. However these constraints were still a fair bit stronger than the constraints required by a true Duistermaat and Guillemin analogue. Additionally, their technique left a persistent constant in the case $n - d = 1$, that appeared superfluous upon empirical testing. In a working paper, I thus proved two results:

First using heat kernel techniques that in the case that $n - d - 1$ if

$$N_H(\lambda) = C_{H,M}\lambda^{n-d} + C + o(1)$$

then $C = 0$.

Second building upon some of their stationary phase result together using microlocal techniques inspired from Ivrii [Ivr80] and Wyman (2017) [Wym17], I was able to prove that under the condition the the set of looping directions has measure 0,

$$N_H(\lambda) = C_{H,M}\lambda^{n-d} + o(1).$$

This condition is laxer and is essentially equivalent to the Duistermaat and Guillemin.

2.3 Ongoing Projects

I have a number of open ended research projects that I am currently persuing in conjunction with my advisor Professor Wyman:

First, most directly pertaining to my previous research, Professor Wyman is convinced that there should be a general Weyl type law for the aformentioned counting functions of $N_A(\lambda)$ for general Fourier integral operators A. In that direction, my advisor believes that I ought to be able to prove the assymptotic in the case A, a finite integral operator, as well as possibly a Duistermaat and Guillemin type refinement.

Second, we are studying "near arithmetic" sequences of eigenvalues. This idea has precedent in existing literature- Duistermaat and Guillemin's paper

[DG75], also studied the case in which the geodesic flows is periodic with the same period T. The calculation involves proving that

$$\sum_{\lambda_j \leq \lambda} (1 - e^{(\beta-\mu)T})\rho(\lambda_j - \lambda) = o(\lambda^{n-1})$$

for an appropriate constant β and thus there must be a concentration of the spectrum about intervals $[nT-\epsilon, nT+\epsilon]$ for n an integer for some appropriately sized $0 < \epsilon < T$. We believe a more general result using similar techniques is possible if there exists a Riemmanian submersion onto an appropriate space.

Third, less directly connected to my other work, I am studying work by Lehec about the possibility of a generalized Brascamp-Lieb type formula. Lehec demonstrated some interesting techniques from SDE theory, specifically a generalized version of Borell's formula, could be used to relatively easily prove the classical Brascamp-Lieb formula [Leh14] as well as its analogue over the sphere [Leh17].

3 Connections to Number Theory

As implied in the terminology, spectral geometry is closely related to analytic number theory and automorphic forms. As the Weyl Law and its variants can essentially be interpreted as an analogue of the Selberg Trace formula and some of the open problems could interesting from a number theoretic perspective. For example, the near arithmetic sequences problem could reveal interesting information on the discrete, non-dihedral spectra of Riemmanian manifold $\Gamma \backslash \mathbb{H}$ for $\Gamma < \mathrm{SL}_2(\mathbb{Z})$ which as far as I am aware would be novel; such information is difficult to access because of the lack of neat explicit forms for non-dihedral forms.

I am personally casually interested in analytic number theory and am very interested if there are interesting connections with the Riemmanian context; in particular there are some interesting joint spectral problems raised from some literature.

References

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